Lab 09
Multiplexer and Demultiplexer

Name:	ID:	Section:

Task a. 4:1 MUX

Complete the truth table provided in Table 9.1. As an example, when the Select line input is $S_{1:0}$=00, output will be the binary number stored in $A_{3:0}$

Table 9.1

S_1	S_0	$A_{3:0}$	$B_{3:0}$	$C_{3:0}$	$D_{3:0}$	$Y_{3:0}$
0	0		X	X	X	
0	1	X		X	X	
1	0	X	X		X	
1	1	X	X	X		

Provide appropriately commented code for designed module

Write a testbench to thoroughly test designed mux_4_1 module.

Attach screenshot of Simulation output- make sure to scale properly for visibility of all case.

Task b. 1:4 DEMUX

Complete the truth table provided in Error: Reference source not found. As an example, when the Select line input is $S_{1:0}$=00, output EnA will be 0 and the rest of the outputs will remain high.

Table 9.4

S_1	S_0	*En*	*EnA*	*EnB*	*EnC*	*EnD*
0	0	0	0	1	1	1
0	1	0				
1	0	0				
1	1	0				
X	X	1	1	1	1	1

Provide appropriately commented code for designed module

Add your testbench here

Attach screenshot of waveform results here

Task c. Top Level Module

Write a Verilog module topLevelModule that combines MUX (**mux_4_1**) and DeMUX (**demux_1_4**) modules created in this lab and binary-hexadecimal decoder module already created in Lab 6, according to the block diagram of Error: Reference source not found. This module's inputs includes four 4-bit data lines (i.e., A[3:0], B[3:0], C[3:0], D[3:0]), 2-bit select line (i.e. S[1:0]) to select from the data for binary-hexadecimal decoder, and a 1-bit active low enable signal (i.e. En). The modules output includes one 7-bit data line for seven segments (i.e., Y [6:0]) and four 1-bit Enable signals (i.e., EnA, EnB, EnC, EnD).

Provide appropriately commented code for your design module

Add snapshot of RTL Schematic of the Top Level Module:

Post Lab Exercise:

a) *Provide appropriately commented code for designed module, code should contain meaningful variable naming.*

b) *Provide appropriately commented code for the testbecnch, code should contain meaningful variable naming.*

c) *Attach screenshot of waveform results here. Make sure the irrelevant area of the snip is cropped.*

Assessment Rubric
Lab 09
Multiplexer and Demultiplexer

Name:	Student ID:

Marks Distribution:

		LR2 Code	LR5 Results	LR9 Report
In - Lab	Task a	10	10	20
	Task b	10	10	
	Task c	10	5	
Exercise		10	5	
Total Marks: 100		/40	/30	/20
CLO Mapping		CLO2	CLO2	CLO4

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission and Viva	/10	CLO 4

CLO	Total Points	Points Obtained
2	70	
4	30	
Total	100	

For description of different levels of the mapped rubrics, please refer to the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Rubrics

#	Assessment Elements	Level 1: Unsatisfactory	Level 2: Developing	Level 3: Good	Level 4: Exemplary
LR2	**Program /Co de/ Simulati on Model/ Network Model**	Program/code/ simulati on model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/ simula tion model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/ simula tion model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/ simulati on /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR5	**Results & Plots**	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR7	**Viva**	Response shows a complete lack of understanding of the assigned / completed task	Response shows shallow understanding of the assigned task	Response shows substantial understanding of the assigned task	Response shows complete understanding of the completed task. The student is able to explain all the related concepts.
LR9	**Report**	No summary provided. The number/amount of tasks completed	Couldn't provide good summary of in-lab tasks. Some major tasks were	Good summary of In-lab tasks. All major tasks completed except	Outstanding Summary of In-Lab tasks. All task completed and

		below the level of satisfaction and/or submitted late	completed but not explained well. Submission on time. Some major plots and figures provided	few minor ones. The work is supported by some decent explanations, Submission on time, All necessary plots, and figures provided	explained well, submitted on time, good presentation of plots and figure with proper label, titles and captions